

OPERATION: Chopper CP-10P

A TRAGEDY IN SEVERAL ACTS

*"The supreme art of war is to subdue the enemy without fighting."
— Sun Tzu, The Art of War
(The enemy being gravity, time, and plastic)*

Presented by: A Soldier Who Has Seen Things

✂ May 2026 ✂

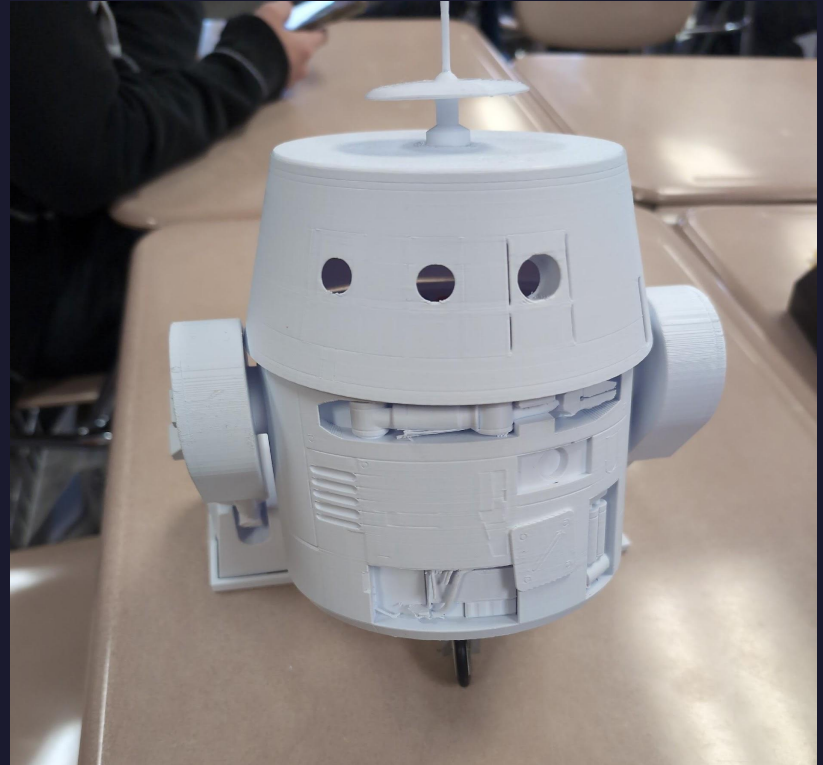
CHAPTER I: THE VISION

(Every great war begins with optimism)

What I promised:

- ✓ An RC droid — fully assembled
- ✓ Motors, servos, tracks, arms
- ✓ Camera with live video feed
- ✓ Obstacle detection
- ✓ Bluetooth/WiFi control
- ✓ A working robot

*"If you know the enemy and know yourself,
you need not fear the result of a hundred battles."
— Sun Tzu (I knew neither)*

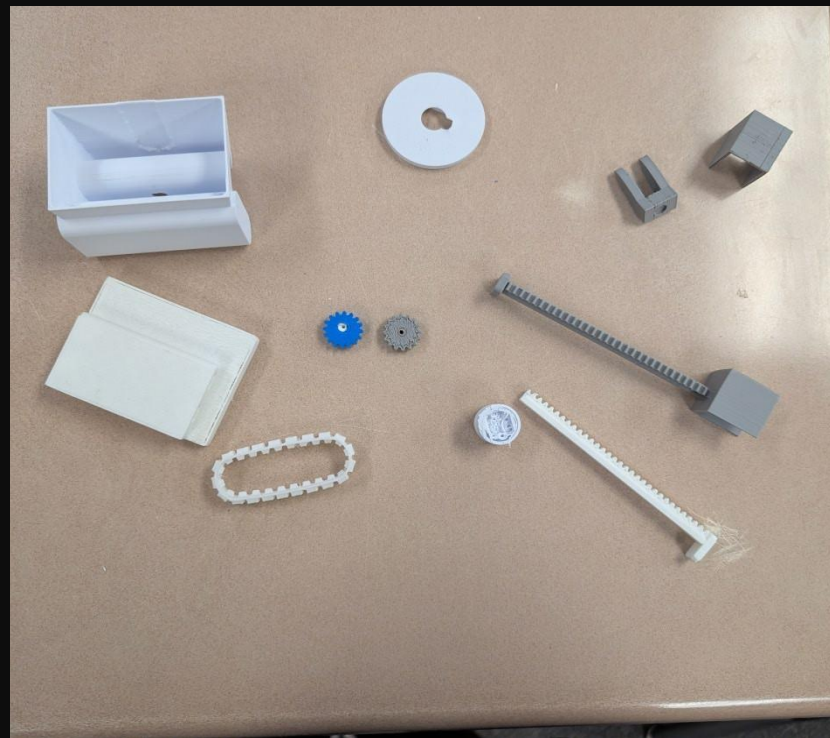


CHAPTER II: THE REALITY

(A brief intermission from reality)

What I actually have:

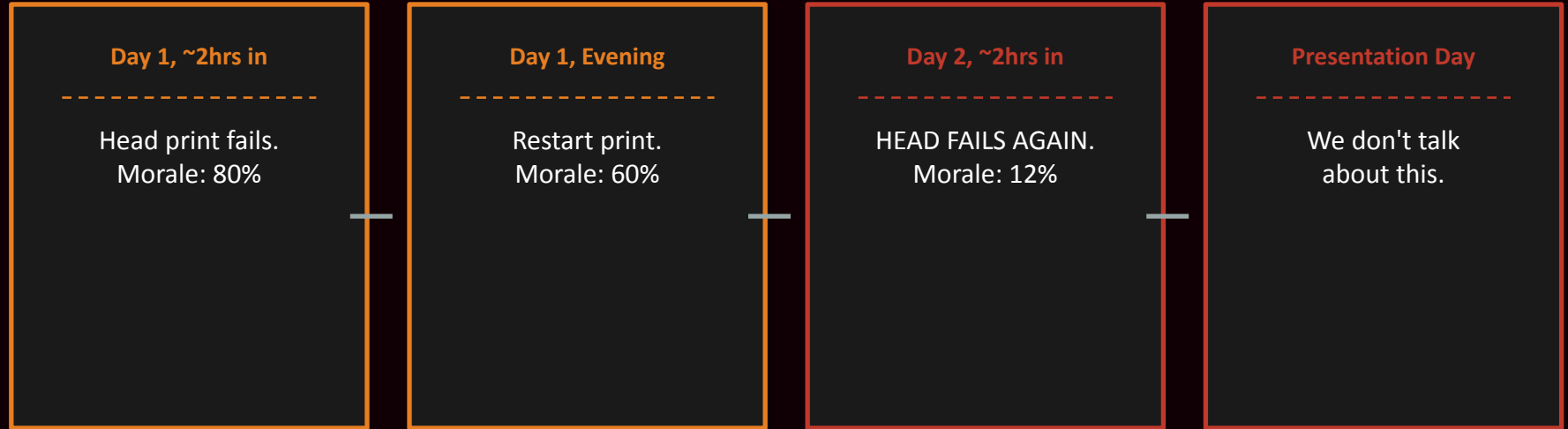
- ✓ Assembled? Kinda.
- ✓ Parts exist (somewhere)
- ✗ Head failed. Twice.
- ✗ Failed at hour 2 of printing. Both times.
- ✓ The code works (allegedly)
- ✗ Cannot demonstrate this claim
- ✓ I have a photo of a robot



The droid failed parts

CHAPTER III: THE HEAD INCIDENTS

(Classified. Rated R. Multiple casualties.)

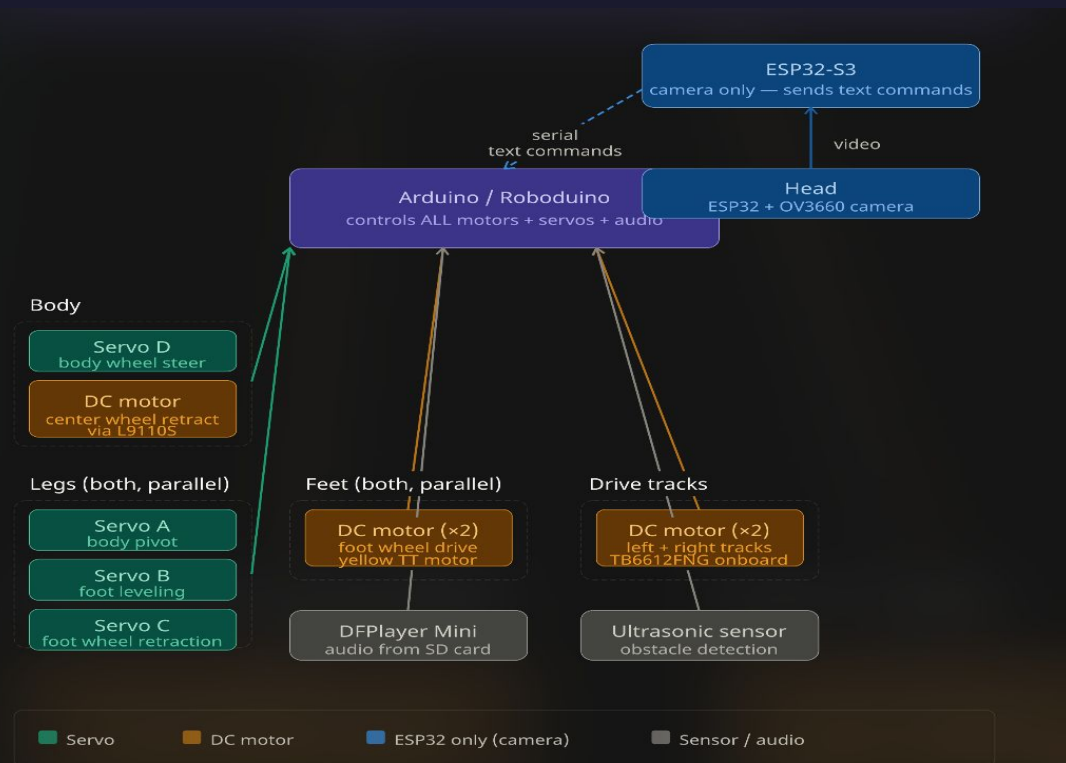


"In the midst of chaos, there is also opportunity."

— Sun Tzu

(The opportunity was to learn that FDM printing hates me personally)

CHAPTER IV: WHAT ACTUALLY WORKS



The Code Architecture

Arduino/Roboduino

Brain of operation.
Controls ALL motors + servos + audio

ESP32-S3 + Head Cam

OV3660 camera → live video
(Head not printed, but conceptually it works)

DFPlayer Mini

Audio from SD card.
For dramatic beeping

Ultrasonic Sensor

Obstacle detection.
So it doesn't crash
(Unlike the print head)

CHAPTER V: MAY 26 2026. (Yesterday)

THINGS GO VERY WRONG.

#1

Head print fails at ~hour 2.
Printer: 'Not today.'

2

Restarted print with optimism.
Optimism: immediately punished.

3

Head print fails AGAIN at ~hour 2.
Same spot. Same failure. Different emotional stage.

4

No head = no camera = no robot.
(Spiritually speaking.)

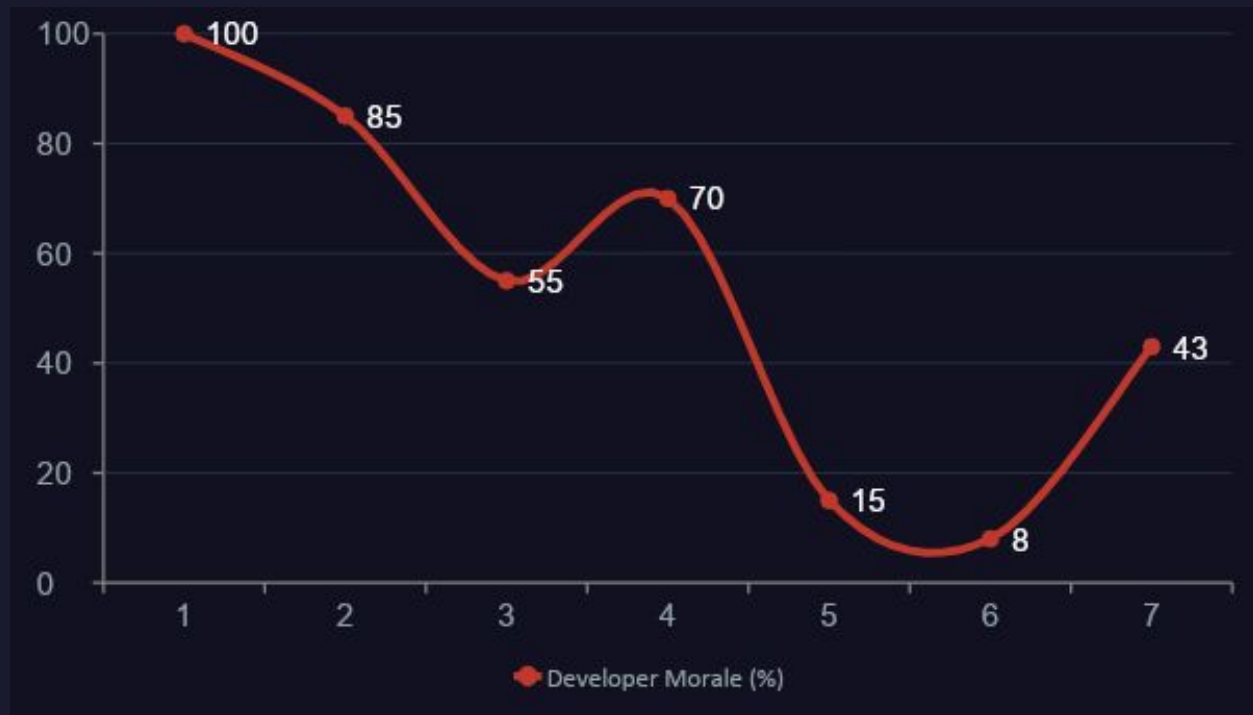
5

The code compiles and runs.
Cannot prove this. Robot not assembled.

6

This slideshow exists.
It is the product. We are the product.

CHAPTER VI: My MORALE REPORT



43%

Reason:
The code still
compiles.

"The general who
advances without
coveting fame...
thinks only of his
country."
— Sun Tzu

(He never had to
3D print anything)

CHAPTER VII: What I would do differently

(A complete accounting of assets, sorted by vibe)

I would not procrastinate and start my project earlier because i couldn't get it fully working in time

I would test my parts before actually using them to make sure they aren't faulty parts

I would also 3d design more thoroughly and less rushed so I don't fail prints due to my designs

Not try and learn blender to make this project with a time crunch

Test first with a breadboard vs just instantly soldering it

Not buy a game during the last final days I'm supposed to be working on it

CHAPTER VIII: THE RUBRIC

(Addressing the elephant in the room. The elephant is my grade.)



Working prototype

It works. It just needs... a body. And a head. And assembly.



Code functionality

100% functional. The Arduino code runs perfectly. Trust me.



System architecture

Fully designed. ESP32-S3, Roboduino, motors, the works.



Documentation

Diagram exists. It's right there in slide 5. Look at it.



Presentation quality

Debatable. But it has Sun Tzu quotes. That's gotta count.



Head is printed

Failed. Twice. At ~hour 2. This is a known issue.

EPILOGUE

The droid is not dead.

It is a work in progress...

*"He will win who knows when to fight and when not to fight."
— Sun Tzu, The Art of War*

(Today, we chose not to fight. Strategically.)

The code compiles. The vision exists. The parts are real.

The head will be printed. Someday. Not today. But someday.

Thank you for listening to the presentation about 3D printer trauma and fails. (I was soldering and coding until 4:30 last night i need sleep)